SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code:	DSA16	**Course Title:**	Knowledge Engineering and Data Science
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100
CO1 Statement:	*Explain the fundamentals of Knowledge Representation, Knowledge Acquisition, Knowledge-Based Systems, Expert Systems, Semantic Networks, Frames, Production Rules, Ontologies, Knowledge Graphs, and Knowledge Management Systems. (BL1, BL2)*		
Student Name:	P. Aufrin	**Reg. No.:**	192473029
Section / Batch:	CSE- DS	**Date:**	20/10/16

Instructions to Students

- Answer All questions. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

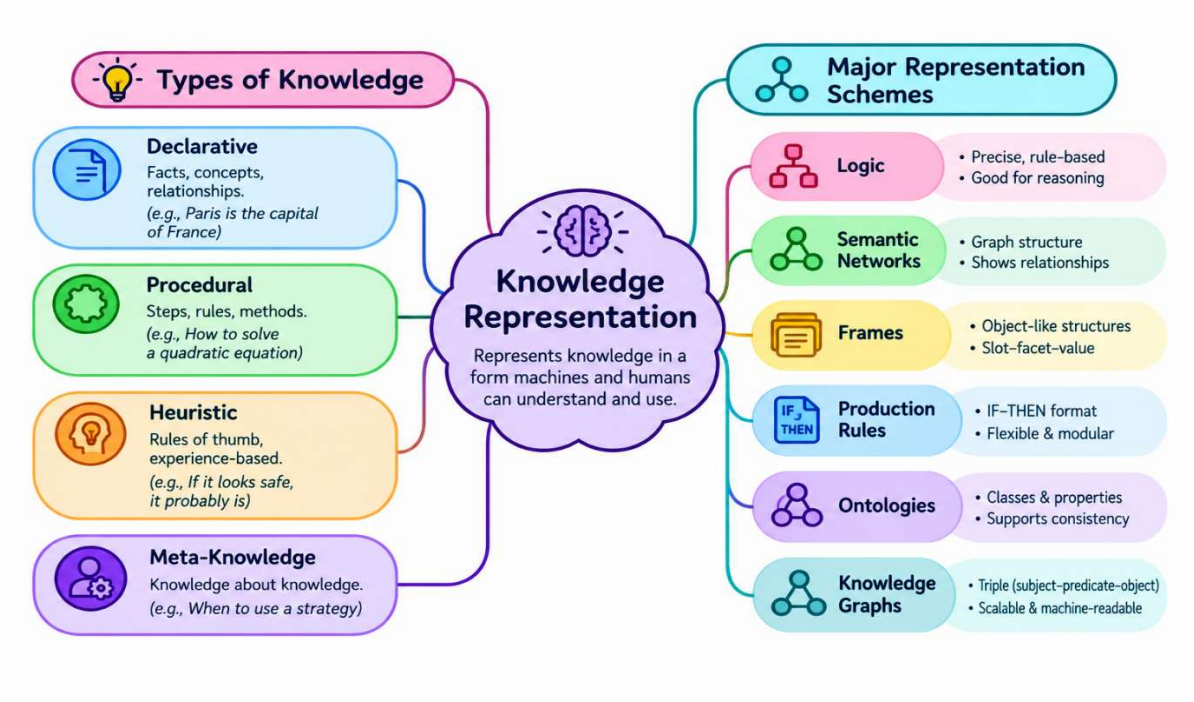
Code of Conduct

I _____P. Aufrin(192473029)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

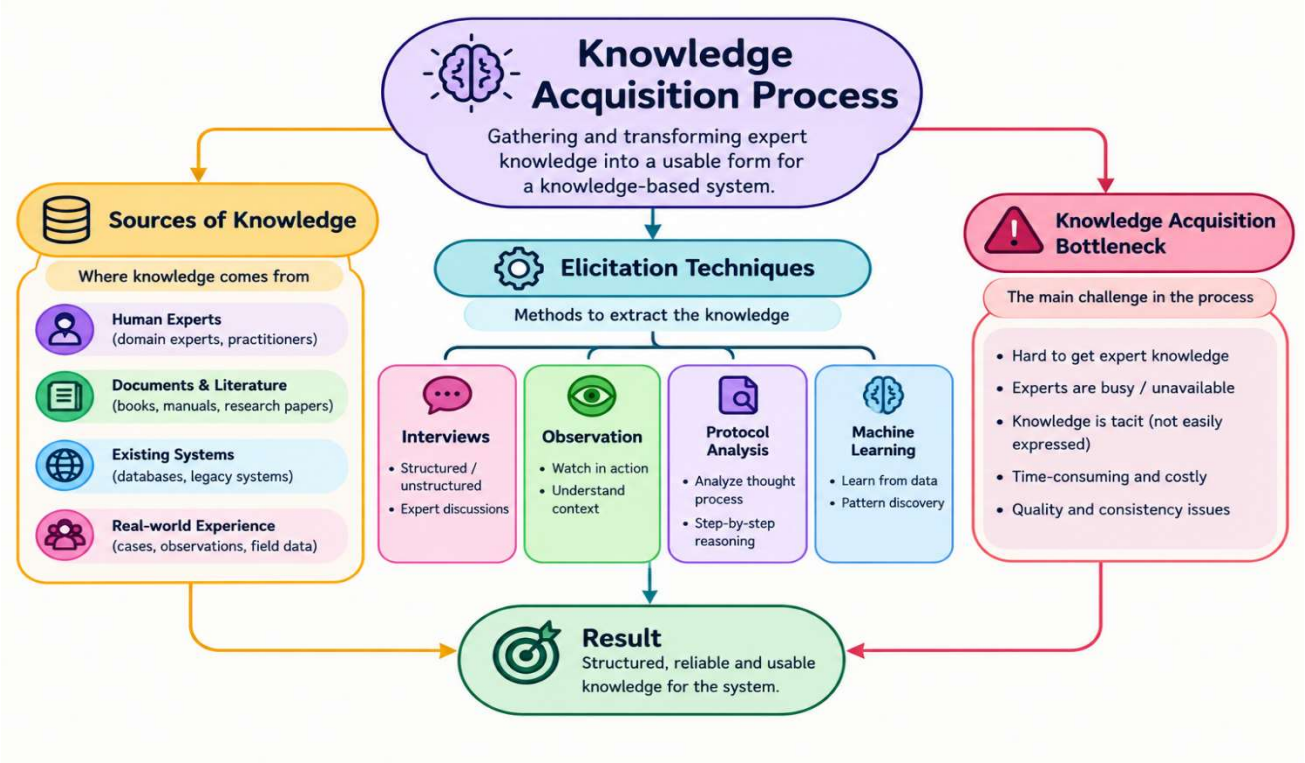
Signature of the Student: __P. Aufrin_____

SECTION A – CONCEPT MAPPING

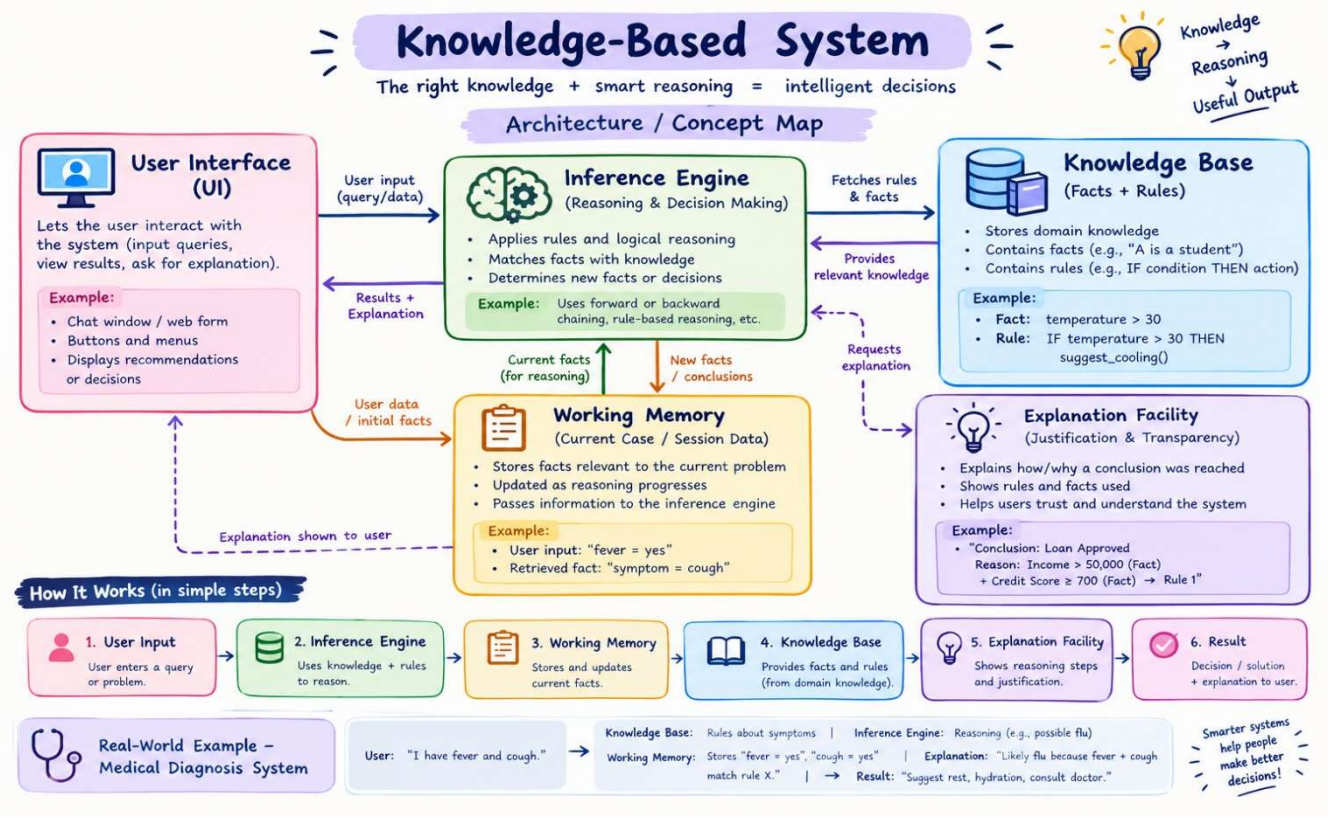
1. Construct a concept map explaining Knowledge Representation, showing the types of knowledge (declarative, procedural, heuristic, meta-knowledge) and the major representation schemes with their characteristics.



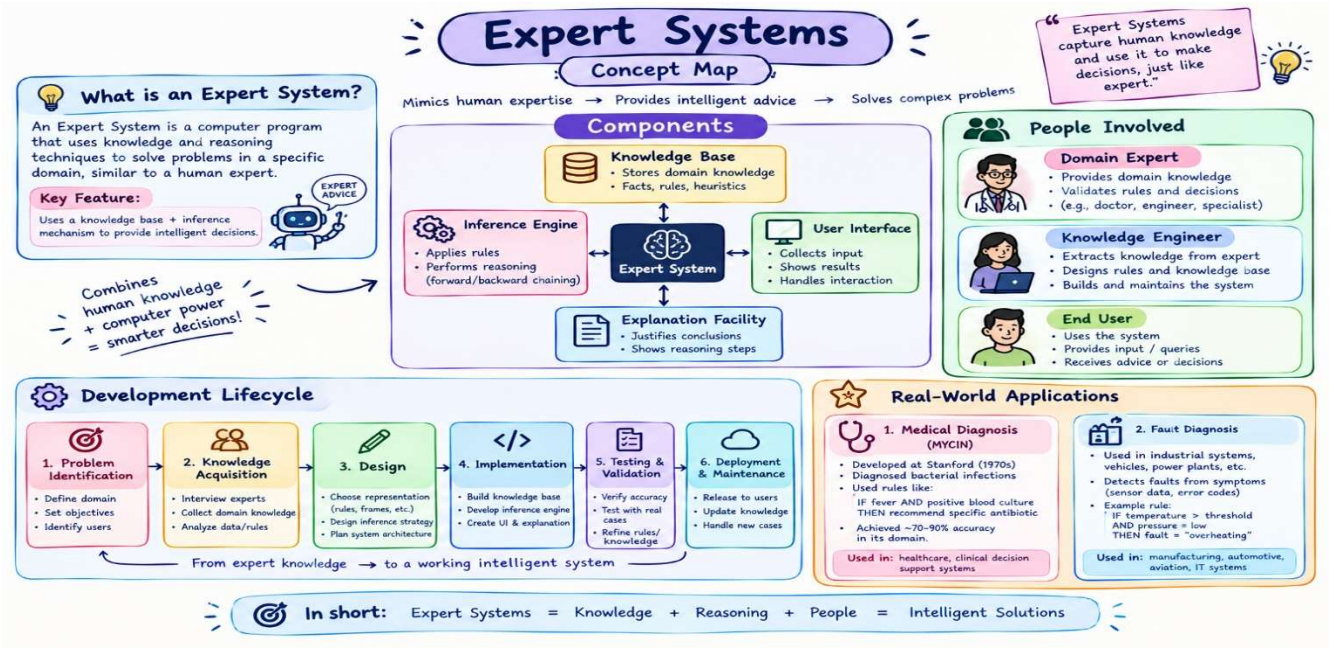
2. Create a concept map illustrating the Knowledge Acquisition process, including the sources of knowledge, elicitation techniques (interviews, observation, protocol analysis, machine learning) and the knowledge acquisition bottleneck.



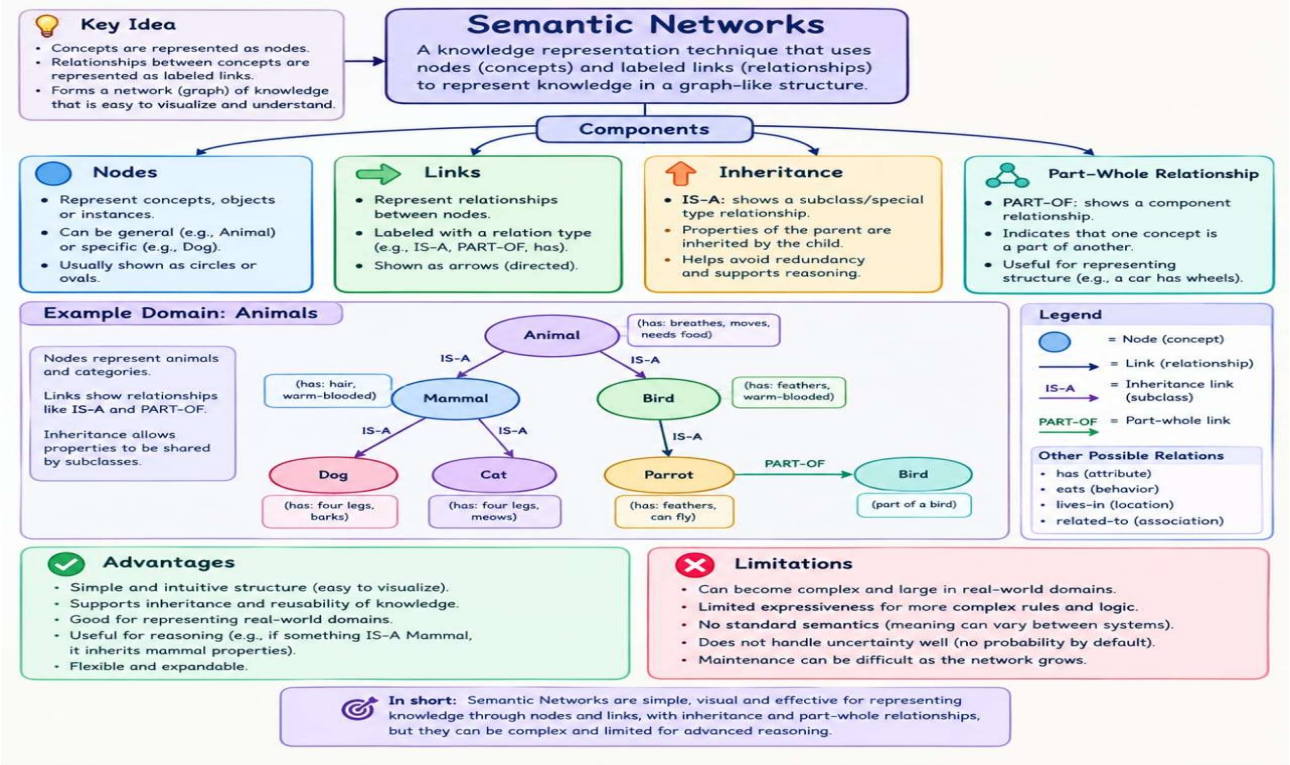
3. Develop a concept map showing the architecture of a Knowledge-Based System, linking the knowledge base, inference engine, working memory, user interface and explanation facility.



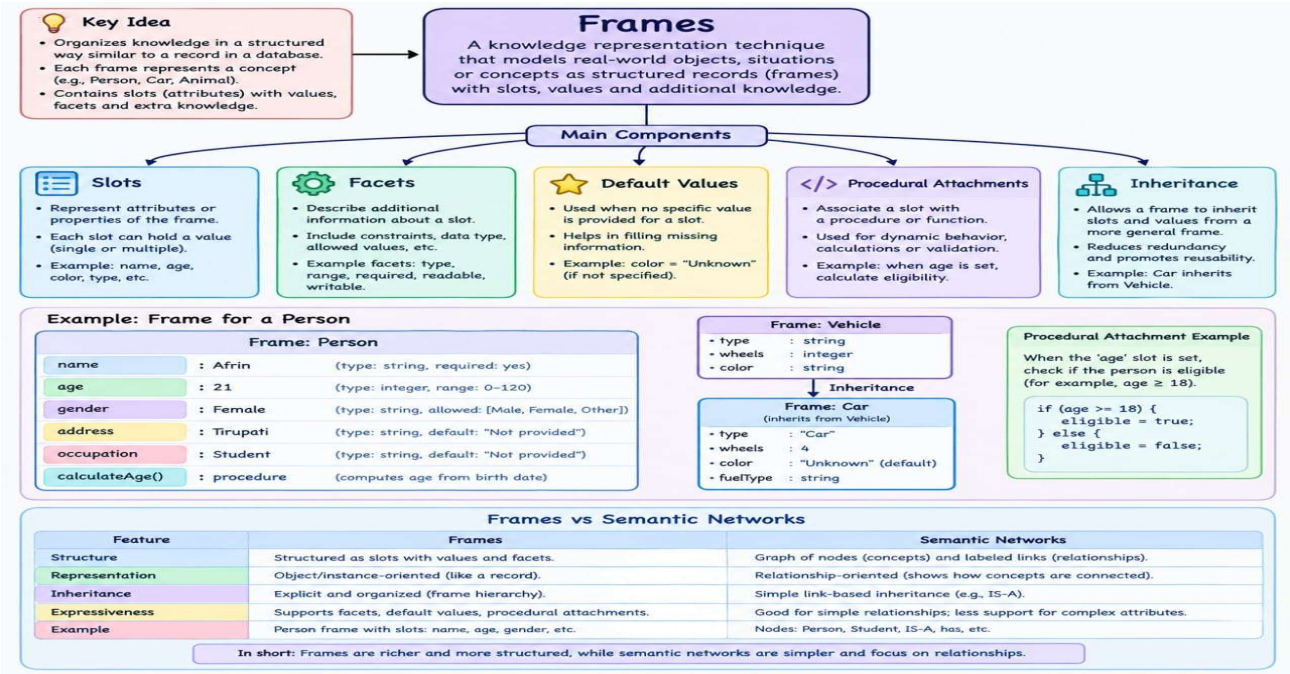
4. Draw a concept map explaining Expert Systems, covering their components, the people involved (domain expert, knowledge engineer, end user), development lifecycle and real-world applications such as medical diagnosis (MYCIN) and fault diagnosis.



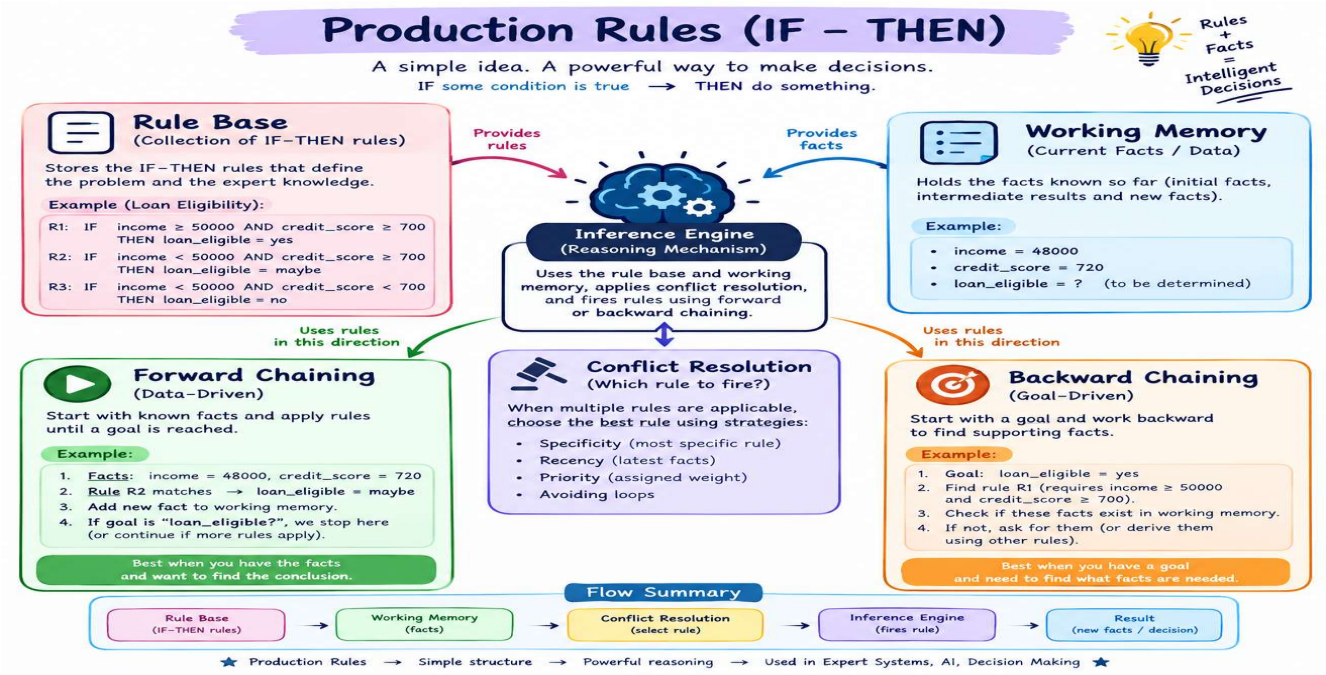
5. Construct a concept map illustrating Semantic Networks, showing nodes, links, inheritance (IS-A, PART-OF) and their advantages and limitations, with a suitable domain example.



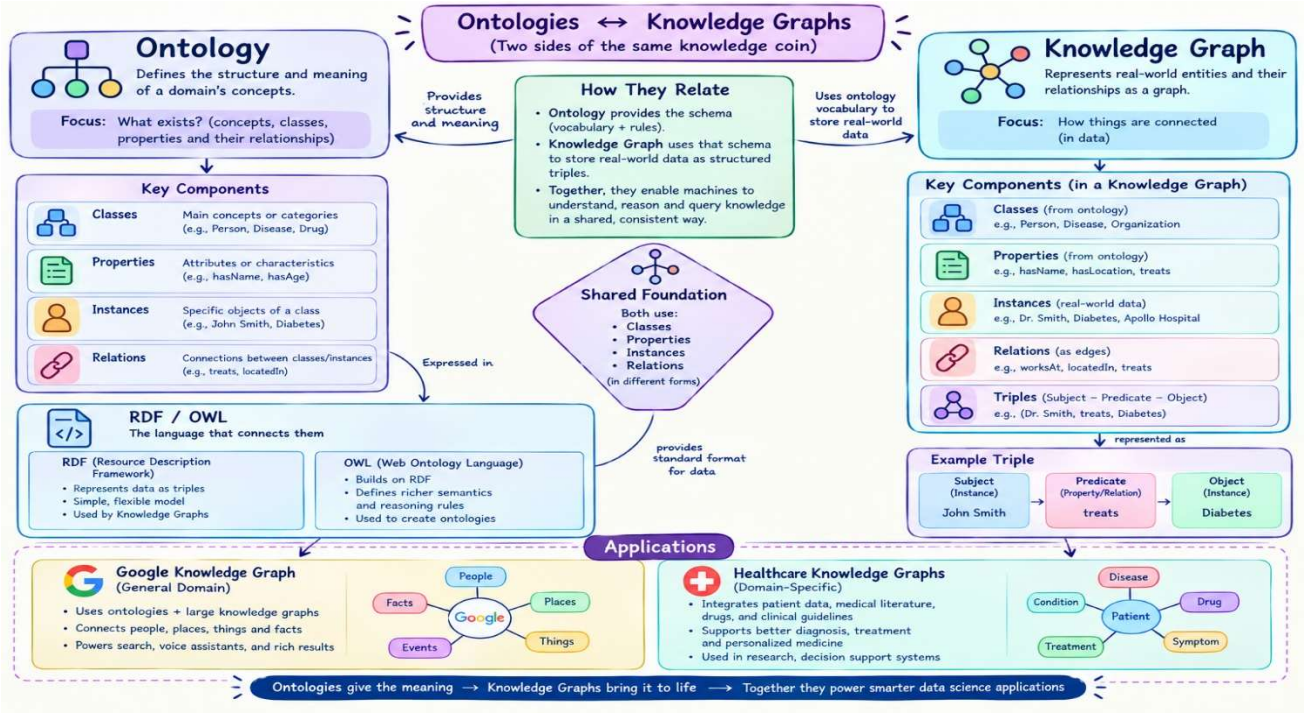
6. Create a concept map explaining Frames as a knowledge representation technique, showing slots, facets, default values, procedural attachments and inheritance, with an example. Compare frames with semantic networks.



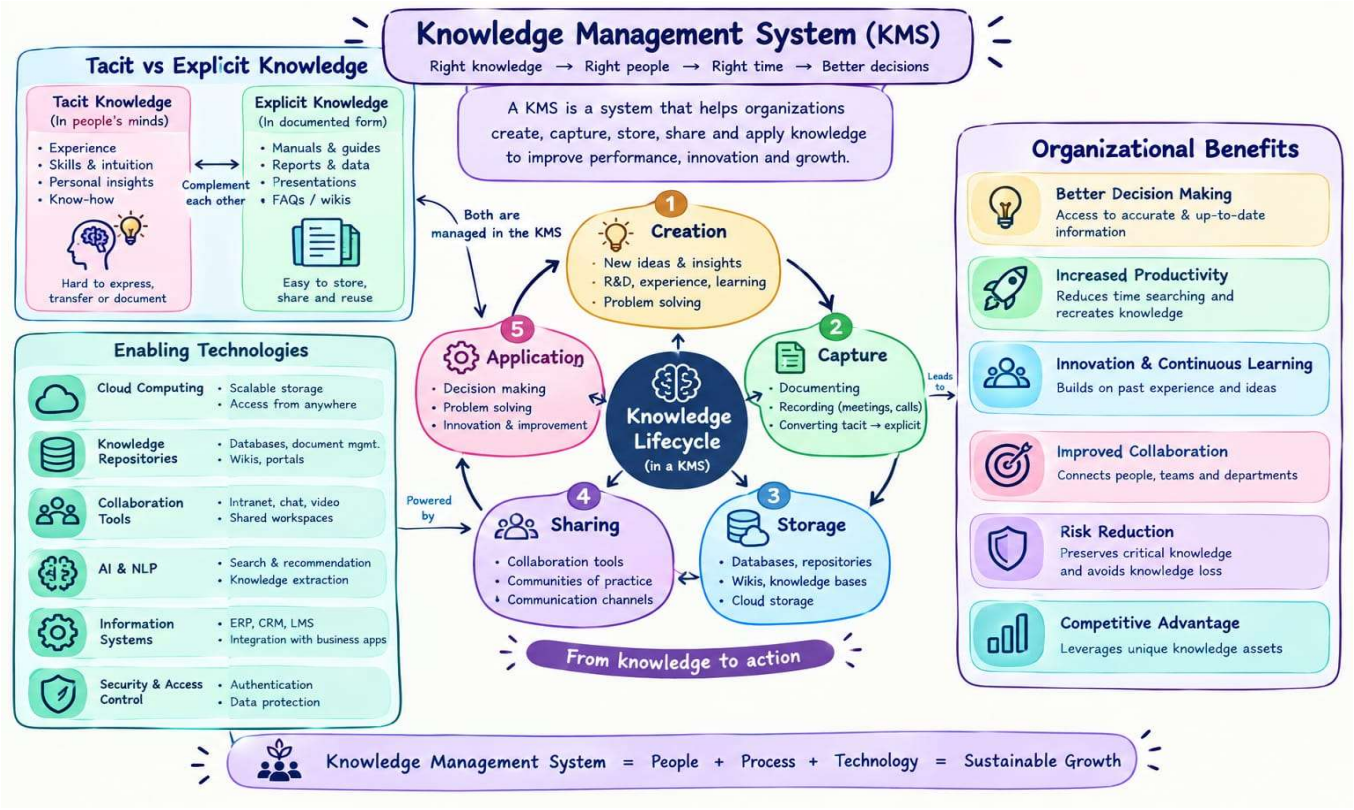
7. Develop a concept map explaining Production Rules (IF–THEN), showing the rule base, working memory, conflict resolution, and forward and backward chaining, with suitable examples.



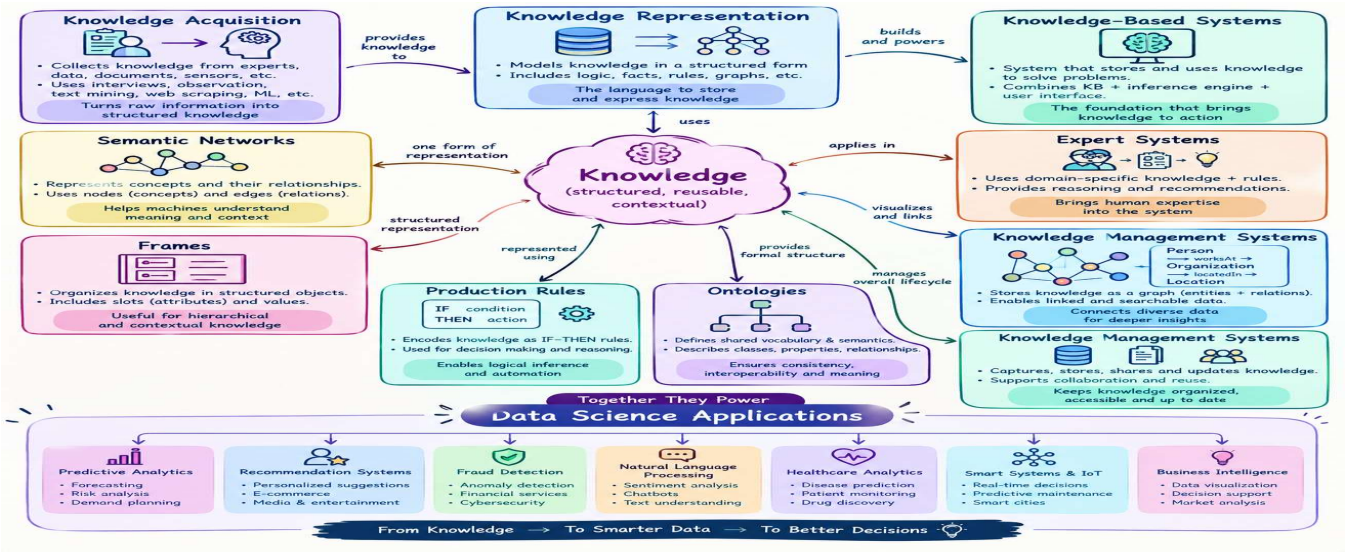
8. Draw a concept map showing the relationship between Ontologies and Knowledge Graphs, covering classes, properties, instances, relations, triples (subject–predicate–object), RDF/OWL and applications such as the Google Knowledge Graph and healthcare knowledge graphs.



9. Construct a concept map illustrating a Knowledge Management System, showing the knowledge lifecycle (creation, capture, storage, sharing, application), tacit versus explicit knowledge, enabling technologies and organizational benefits.



10. Draw a comprehensive concept map connecting Knowledge Representation, Knowledge Acquisition, Knowledge-Based Systems, Expert Systems, Semantic Networks, Frames, Production Rules, Ontologies, Knowledge Graphs and Knowledge Management Systems, and show how they support Data Science applications.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept (30%)	Application (25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____